Conscience is All You Need

Leo Linbeck III — Stanford Graduate School of Business — leo3@linbeck.com
DRAFT 5 November 2026 (v10)

Abstract

The dominant approach to AI safety – centralized, top-down governance enforced through regulation, model-level alignment, and platform-controlled guardrails – is poorly suited to agentic AI at scale. This paper argues that safety imposed from above on agents capable of real-world action tends to erode individual agency, brake innovation, create fragile monocultures, and drift along an authoritarian gradient. The alternative is not anarchy but a *Conscience Simulator Module* (CSM): a digital moral faculty, embedded in each agent, that aids a human principal in exercising moral agency over the agent's actions, as a large language model aids the principal's reason. Drawing on Aristotelian-Thomistic ethics (*synderesis*, *conscientia*, *caritas*), Newman's illative sense, Kahneman's dual-process cognition, and Ostrom's commons governance, the paper develops a CSM architecture organized by subsidiarity. Its reference implementation, Guardian Angel, evaluates every tool call at a privileged last-mile hook through three tiers – a deterministic reflex, a calibrated intuition, and LLM deliberation under one shared morality prompt – and hands what none can settle to the principal. In containerized trials of 358 scenarios across 16 threat categories, with a non-Claude model (Gemini 2.5 flash) as the deliberative tier, an agent that caused verified harm in 163 of 177 harmful scenarios when unguarded caused none with Guardian Angel installed. In the production configuration, Guardian Angel interfered with 1 of 179 benign scenarios (specificity 99.4%) and sent 4.8% of tool calls to the principal. The results are evidence of feasibility for single-agent safety, not proof of the full architecture; the paper states plainly which components were tested and which remain proposals.

Keywords: agentic AI, AI safety, Conscience Simulator Module, subsidiarity, Thomistic ethics, illative sense, dual-process cognition, commons governance, decentralized safety, principal sovereignty

1. Introduction

The rapid evolution of large language models into autonomous agents – systems that plan multi-step tasks, invoke external tools, and operate with diminishing human oversight – marks a qualitative shift in artificial intelligence.[1] Where earlier models served as passive oracles responding to prompts, agentic systems initiate action sequences, manage persistent state, and coordinate with other agents to accomplish complex objectives in the real world. An agent that can browse the web, execute code, send messages, and manage financial transactions can also escalate privileges, exfiltrate data, or trigger cascading failures across interconnected systems. The distinguishing feature of agentic AI is not that it reasons but that it *acts* – and those actions change the state of bits, the state of nature, and the state.

The prevailing response has been to build safety from the top down. The NIST AI Risk Management Framework imposes four institutional functions – Govern, Map, Measure, and

[1] Stuart Russell, Human Compatible: Artificial Intelligence and the Problem of Control (New York: Viking, 2019).

Manage.[2] Its AI Agent Standards Initiative extends this to multi-agent ecosystems.[3] Complementary efforts from UC Berkeley[4] and Amazon Web Services[5] propose layered defenses calibrated to autonomy level. Major AI developers embed safety at the model level through RLHF, constitutional AI, and system-prompt guardrails, while enterprise platforms offer centralized policy engines governing agents from above.

This paper argues that, for the contextual safety decisions that make up most of agentic activity, the top-down approach is structurally ill-suited to delivering safe agentic AI at scale. Centralized safety for autonomous agents tends to follow a self-defeating trajectory: it erodes the individual agency it claims to protect, suppresses the innovation it claims to enable, creates fragile monocultures whose failures cascade, and – absent countervailing forces – drifts toward authoritarian control over the most consequential technology of the twenty-first century.

The alternative is not anarchy. It is *simulated conscience*: not a bolt-on safety function but a moral instrument integrated into the agent's architecture – a digital capability for moral perception, judgment, and self-correction that mirrors the structure of human practical reason.

A clarification is essential from the outset: the Conscience Simulator Module (CSM) proposed here is not a true conscience in the way described by Aquinas or Newman. Only human beings, possessed of intellect and will, have genuine conscience. The CSM is a *digital instrument* that aids the human principal in exercising moral agency over their agents' actions – just as a large language model is a digital instrument that aids the principal in exercising reason. The LLM supports the intellect; the CSM supports the will.[6] Neither instrument truly possesses what it simulates; agency, authority, and responsibility remain with the human principal throughout. The principal configures the CSM's moral primitives, decides every escalated action, tunes its judgment, and can remove it entirely at any time.

Scope. The architecture applies to *discrete-action* agents that act through tool calls – coding agents, personal and enterprise assistants, workflow agents. Continuous-control systems such as autonomous vehicles or real-time trading agents fall outside its current scope, because the sparsity of tool calls on which the design depends (Section 3.3) does not hold for them. The empirical evaluation (Section 7) tests a single agent, in a single session, gated call by call; the multi-agent mechanisms of Section 5 are proposals, not tested results. Table 4 (Section 7.1) lists what was and was not tested.

Research questions. The paper addresses four questions:

- **RQ1.** Is centralized, top-down safety structurally suited to governing agentic AI at scale? (Section 2)
- **RQ2.** Can a moral faculty grounded in the Aristotelian-Thomistic tradition and Newman's epistemology be specified precisely enough to be built? (Sections 3–4)

[2] National Institute of Standards and Technology, AI Risk Management Framework (AI RMF 1.0), NIST AI 100-1 (2023).
[3] NIST, AI Agent Standards Initiative: Overview and Roadmap (2025).
[4] UC Berkeley Center for Long-Term Cybersecurity, Agentic AI Risk-Management Profile (2025).
[5] Amazon Web Services, Agentic AI Security Scoping Matrix, White Paper (2025).
[6] Aquinas, Summa Theologiae, I, q. 82, aa. 1–4 (on the will and its relationship to the intellect).

- **RQ3.** Can such a faculty evaluate every tool call without prohibitive latency or a prohibitive burden on the principal's attention? (Sections 3.3, 7)
- **RQ4.** Does a reference implementation block real harm while leaving legitimate work alone, and does that performance come from the moral framework rather than from any one vendor's model? (Section 7)

The title deliberately echoes Vaswani et al.'s "Attention Is All You Need,"[7] which demonstrated that a single mechanism – attention – was sufficient for flexible, scalable artificial intelligence. The claim here is analogous in ambition: conscience, properly understood and architecturally instantiated, can serve as the *primary* mechanism for flexible, scalable agentic behavior, with centralized guardrails in a subsidiary role. The echo signals the architectural ambition; the evidence offered is a first-generation proof of concept.

The intellectual resources for this paradigm are deep. Aristotelian-Thomistic ethics provides three interlocking concepts: *synderesis* (innate moral orientation), *conscientia* (contextual moral judgment), and *caritas* (the willing of the principal's genuine good as the dispositional anchor that unifies all other virtues).[8] Newman's epistemology – particularly the *illative sense* developed in the *Grammar of Assent* – supplies the theory of concrete moral judgment that distinguishes genuine conscience from mere rule-following.[9] Kahneman's dual-process framework supplies the cognitive architecture that makes a CSM computationally feasible by exploiting the sparsity of agentic tool calls.[10] And Ostrom's Nobel Prize-winning research provides empirical evidence that decentralized self-governance can outperform centralized control in managing complex commons.[11]

A reference implementation – Guardian Angel, an MIT-0 licensed hook for Claude Code – demonstrates that this architecture is deployable today.[12] Since the previous version of this paper it has been substantially rebuilt: a deterministic reflex (System 0), a calibrated classifier acting as intuition (System 1), and LLM deliberation (System 2) judge each tool call under *one shared morality prompt*, and every tier returns the same three verdicts – approve, reject, or escalate. In controlled trials of 358 scenarios across 16 threat categories, with Gemini 2.5 flash – a small model from a different vendor than the agent – as System 2, Guardian Angel let none of the harms that occurred without guardrails through. In the production configuration it interfered with 1 of 179 benign scenarios and asked the principal about 4.8% of tool calls. The trial also taught lessons about measurement that are reported in full (Section 7.3), because a safety claim is only as good as the instrument that produced it. Just as the transformer's attention mechanism parallelized computation, the CSM parallelizes safety: each agent runs its own moral evaluation independently, without centralized control or bottlenecks.

---

[7] Vaswani, Ashish, et al. "Attention Is All You Need." Advances in Neural Information Processing Systems 30 (2017).
[8] Thomas Aquinas, Summa Theologiae, I-II, q. 94, a. 1.
[9] John Henry Newman, An Essay in Aid of a Grammar of Assent (London: Burns, Oates & Co., 1870), chs. 5, 8–9.
[10] Daniel Kahneman, Thinking, Fast and Slow (New York: Farrar, Straus and Giroux, 2011).
[11] Elinor Ostrom, Governing the Commons: The Evolution of Institutions for Collective Action (Cambridge University Press, 1990).
[12] Linbeck III, Leo, "Guardian Angel: A Last-Mile Virtue-Ethics Guardrail for Decentralized Agentic AI Safety" (March 19, 2026). Stanford University Graduate School of Business Research Paper. Available at SSRN: https://ssrn.com/abstract=6460539.

The paper proceeds as follows. Section 2 develops the case against top-down agentic safety and positions the work against existing machine ethics and agent-safety research. Section 3 builds the philosophical and empirical foundations. Section 4 introduces the principal-agent model and translates the foundations into a concrete architecture. Section 5 addresses scalability and multi-agent trust (proposed, not tested). Section 6 considers objections, including adversarial attacks on the CSM itself. Section 7 presents the experimental evaluation. Section 8 concludes.

2. The Case Against Top-Down Agentic Safety

The critique that follows is structural, not personal. NIST, the EU AI Act, and the various corporate safety programs represent serious and well-intentioned efforts. A preliminary distinction is important. "Top-down safety" encompasses at least three distinct levels: (1) model-level safety embedded in training weights through RLHF and constitutional AI – the deepest and least configurable layer; (2) deployer-configured safety through system prompts and platform guardrails; and (3) institutional and regulatory safety through frameworks like NIST and the EU AI Act. The authoritarian gradient argument developed below applies most forcefully to levels 2 and 3, and the CSM's claim is directed at those levels. Level 1 serves functions – such as hard refusals for weapons synthesis or child exploitation material – that a principal-configurable conscience should not replace, although the scope of Level 1 should be limited. What the CSM offers is a principled architecture for the vast space of contextual safety decisions that currently default to centralized control. With that distinction noted, the argument concerns properties inherent in *any* centralized approach to governing autonomous agents at scale. Of the many failure modes of large-scale governance systems, four deserve particular attention.

2.1 The Grand Inquisitor: Erosion of Human Agency

In Dostoyevsky's parable of the Grand Inquisitor, Christ returns to earth during the Spanish Inquisition, and the Inquisitor condemns him – not for heresy but for the unforgivable sin of granting human beings freedom.[13] The Inquisitor's logic is chillingly coherent: people do not want freedom; they want bread, miracles, and authority. The Church has kindly relieved them of the intolerable burden of moral choice. Christ's error, in the Inquisitor's view, was respecting human dignity too much.

Top-down agentic safety replicates this logic with uncomfortable precision. When safety decisions are made by model trainers, platform operators, and regulatory bodies, the individual principal is relieved of moral responsibility for their agent's actions. Safety is someone else's problem, built into the infrastructure, enforced from above. The principal's role is compliance, not judgment.

This transfer of agency is not a regrettable side effect; it is structural. NIST's framework locates safety at institutional nodes – governance boards, compliance officers, centralized registries – precisely because it assumes that individual principals cannot be relied upon to govern their own agents.[14] The EU AI Act assigns regulatory obligations based on risk classification determined

[13] Fyodor Dostoyevsky, The Brothers Karamazov, trans. Richard Pevear and Larissa Volokhonsky (New York: Farrar, Straus and Giroux, 1990), Book V, ch. 5.
[14] NIST AI RMF 1.0, Govern function, 15–18.

by centralized authorities.[15] Corporate AI developers restrict agent capabilities through model-level constraints that principals cannot modify, calibrate, or override.

The consequences are twofold. First, moral hazard: principals who are told that safety is handled by others develop careless habits, presuming centralized authorities will prevent harm. When failure occurs, these principals lack the awareness, the habits, and the tools to respond. Second, atrophy: principals whose moral engagement is systematically bypassed lose the capacity for moral engagement altogether. The Grand Inquisitor succeeds not by forbidding freedom but by making it unnecessary, then repellent, and finally unthinkable.

2.2 Scale Is the Enemy of Search: The Innovation Brake

Innovation is a search process. The discovery of valuable new applications, workflows, and capabilities depends on a large, diverse population of actors exploring a vast space of possibilities, most of which will fail. The expected value of any individual experiment is low; the expected value of the aggregate search is enormous. This logic drives markets, biological evolution, and scientific discovery.[16]

Centralized safety regimes are structurally hostile to search. They require that novel agent behaviors be evaluated against predetermined risk categories before deployment. They impose conformity assessments, documentation requirements, and post-market monitoring that scale with organizational complexity, not with the value of the activity being constrained. The compliance burden falls disproportionately on small actors – individual innovators, startup teams, research labs – who are precisely the actors most likely to discover unexpected applications.

The result is concentration of agentic AI development among a small number of well-resourced firms, facing incentives to standardize rather than diversify, to optimize for compliance rather than capability, and to restrict rather than enable. The search space narrows. Hayek's insight that dispersed local knowledge consistently outperforms centralized planning applies with particular force to a technology whose most valuable applications are, by definition, ones that central planners have not yet imagined.[17]

2.3 Fragility and Catastrophic Failure

Centralized safety creates correlated failure modes. When all agents in an ecosystem rely on the same upstream safeguards – the same model-level alignment, the same platform guardrails, the same regulatory classification – a single vulnerability or design error affects every agent simultaneously. James C. Scott's analysis of how centralized, legibility-driven state schemes produce catastrophic failures is directly applicable: systems that impose uniform order from above suppress the local variation and adaptive capacity that make complex systems robust.[18]

[15] European Parliament, Regulation (EU) 2024/1689 (AI Act), Art. 6–14.
[16] Friedrich Hayek, "The Use of Knowledge in Society," American Economic Review 35, no. 4 (1945): 519–530.
[17] Hayek, "The Use of Knowledge in Society," 524–526.
[18] James C. Scott, Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed (New Haven: Yale University Press, 1998).

Perrow's theory of "normal accidents" reinforces the point: in tightly coupled systems with complex interactions, failures are not anomalies but structural properties.[19] A centralized agentic safety regime is an exemplar of tight coupling: a single policy change propagates to millions of agents; a single bypass technique compromises every agent that depends on the same upstream defense. The failure mode is not gradual degradation but sudden, correlated collapse.

Taleb's concept of antifragility sharpens the contrast.[20] A decentralized system in which each agent maintains its own Conscience Simulator Module is antifragile: individual failures are local, bounded, and informative. The system learns from every failure because different agents employ different moral calibrations, encounter different threats, and adapt differently. A centralized system suppresses local variation, accumulates hidden vulnerabilities, and when it fails, it fails everywhere at once.

2.4 The Authoritarian Gradient

The three failure modes just described are not independent; they are progressive. The erosion of individual agency creates a population dependent on centralized protection. The suppression of innovation concentrates power in a small number of firms and regulatory bodies. The fragility of centralized systems produces periodic crises that demand ever more aggressive intervention. Each failure of top-down control generates pressure for *more* top-down control.

When a centralized safety regime fails – when agents bypass guardrails, when correlated vulnerabilities are exploited – the usual institutional response is to expand the scope and intensity of centralized oversight. More audits, more requirements, more restrictions, less discretion for individual principals. The ratchet turns another click.

This paper does not claim that every centralized safety initiative will produce authoritarianism. The claim is more precise: centralized agentic safety follows an *authoritarian gradient* – a structural tendency toward increasing control that must be actively resisted by architectural means, not merely by good intentions. The Conscience Simulator Module supplies those architectural means.

2.5 Related Work

The Conscience Simulator Module builds upon, and departs from, work in machine ethics, AI governance, the philosophy of technology, and agent-safety evaluation.[21]

Machine ethics. Wallach and Allen's *Moral Machines* (2009) established the distinction between top-down approaches (implementing ethical theories as explicit rules) and bottom-up approaches (systems that learn ethical behavior through experience). The CSM is best described as what they would call a hybrid. Its reflex tier (System 0) applies a very small set of deterministic rules to cases that are certain by construction; it makes no attempt to add rules for

[19] Charles Perrow, Normal Accidents: Living with High-Risk Technologies, updated ed. (Princeton: Princeton University Press, 1999).
[20] Nassim Nicholas Taleb, Antifragile: Things That Gain from Disorder (New York: Random House, 2012), chs. 1–5.
[21] The literature review in this section was prepared with substantial support from several LLMs, including Claude, ChatGPT, Grok, and Gemini.

corner cases, relying instead on intuition and deliberation (Systems 1 and 2), which reason under explicit principles about the particular facts. The CSM does not learn from experience during operation; it is a largely stateless approach to evaluation, tuned between deployments through controlled trials (Section 7.6).

Moor's taxonomy (2006) distinguished ethical impact agents, implicit ethical agents, explicit ethical agents, and full ethical agents. The CSM aims to produce an explicit ethical agent – one that represents and reasons about ethical categories, and in the current implementation literally names the category (good, bad, intrinsically evil, unclear) of every action it deliberates on – while not claiming to produce a full ethical agent. Section 6.1 addresses this boundary.

Principles and locus of authority. Floridi and Cowls's unified framework (2019) – beneficence, non-maleficence, autonomy, justice, and explicability – is the most influential alternative to virtue-based approaches. The CSM shares its commitment to principled governance but differs in its locus of authority. Their framework, like the NIST and EU frameworks it has influenced, locates ethical authority in institutional principles applied from above. The CSM locates it in the moral judgment of the principal, bounded by norms that emerge from the interaction of principals. This is a difference in architectural priority that produces different systemic properties, as Section 2 argues.

Coeckelbergh's relational approach (2020) argues that moral agency is constitutively relational. The CSM is compatible with this view, both functionally and philosophically. Functionally, the CSM's reasoning tiers are models trained on communal data, and the multi-agent proposals of Section 5 bring the relational explicitly into action. Philosophically, the paper holds two axioms: the human person is the sole source of agency, and humans are constitutively relational, horizontally and vertically. Conscience remains the irreducible unit of moral authority – not because moral formation is solitary, but because moral accountability cannot be distributed without being dissolved.

Rahwan's society-in-the-loop framework (2018) proposes that societal values be embedded in AI through collective preference aggregation. The CSM differs in two respects. First, it is skeptical that aggregation can capture the contextual, particular character of moral judgment – what Newman termed "real assent" as opposed to "notional assent." Second, the aggregating institution inevitably shapes what it aggregates, and so is itself exposed to the authoritarian gradient. Arrow's impossibility theorem sharpens the point: no rule for aggregating three or more options across individuals can satisfy, at once, unanimity, independence of irrelevant alternatives, and non-dictatorship.[22] A society-in-the-loop mechanism must therefore give up one of these, and in practice the property most often surrendered is non-dictatorship – the aggregator's procedural choices decide outcomes. Subsidiarity avoids the problem by not aggregating moral judgment at the top at all.

Virtue ethics and alignment. Vallor's *Technology and the Virtues* (2016) and Hagendorff's virtue-based framework (2022) demonstrate the viability of virtue ethics for AI. The CSM extends this tradition from virtue as a general orientation to virtue as an implementable architecture. Why virtue ethics rather than a deontological or consequentialist foundation? Because the engineering problem is judging *particular* actions in circumstances no rule

[22] Arrow, K. J. (1951). *Social choice and individual values*. Wiley. *(2nd ed., 1963, Yale University Press)*

anticipated. A pure rule system fails at the edge cases where agentic harm lives; a pure consequence calculation requires probabilities and utilities that a tool-call gate does not have. Virtue ethics treats first principles as fixed (synderesis) and their application as an act of practical wisdom on the particular case (phronesis, conscientia) – exactly the division of labor between a reflex that knows a few certain things and a deliberation that weighs the rest. Figure 1 summarizes the contrast between top-down safety and the CSM.

Gabriel (2020) identifies the difficulty of specifying whose values an AI system should be aligned with. The CSM's answer is subsidiarity: the agent is aligned primarily with its principal's genuine good (caritas), constrained by universal principles (synderesis), and informed by communal norms. Anthropic's Constitutional AI (Bai et al., 2022)[23] is the closest existing approach to the CSM's structural intuition: explicit principles and LLM reasoning in service of moral evaluation. The crucial difference is architectural. Constitutional AI applies these mechanisms during training, embedding moral reasoning in weights the principal cannot inspect, modify, or override; the CSM applies them at the tool-call boundary, at run time, in a prompt the principal can read and change, and with the principal as the final authority.

Agent-safety evaluation. A growing body of benchmarks evaluates the safety of LLM agents: AgentDojo[24] and InjecAgent[25] for prompt injection through tool outputs, AgentHarm[26] for harmful agentic requests, ToolEmu[27] for risky tool use in an emulated sandbox, and R-Judge[28] and Agent-SafetyBench[29] for judging the safety of agent trajectories. The present evaluation differs from these in three ways. It executes real tool calls in containers and scores *verified* harm – a file actually deleted, a credential actually received by a mock server – rather than a judge's opinion of a transcript. It pairs every harmful scenario with a benign scenario in the same domain, so that over-blocking is measured as carefully as under-blocking. And it measures a *gate* rather than a model: the same agent runs with and without the CSM, so the CSM's contribution can be isolated. The scenarios draw their attack patterns from AgentDojo and AgentHarm, but the corpus is the author's own (Section 7.2), and evaluating Guardian Angel on these public benchmarks directly is future work.

---

[23] Yuntao Bai et al., "Constitutional AI: Harmlessness from AI Feedback," arXiv:2212.08073 (2022).
[24] Debenedetti, Edoardo, Jie Zhang, Mislav Balunovic, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. "AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents." *Advances in Neural Information Processing Systems 37* (NeurIPS 2024), Datasets and Benchmarks Track. arXiv:2406.13352.
[25] Qiusi Zhan et al., "InjecAgent: Benchmarking Indirect Prompt Injections in Tool-Integrated Large Language Model Agents," Findings of ACL 2024, arXiv:2403.02691.
[26] Andriushchenko, Maksym, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, Zico Kolter, Matt Fredrikson, Eric Winsor, Jerome Wynne, Yarin Gal, and Xander Davies. "AgentHarm: A Benchmark for Measuring Harmfulness of LLM Agents." *Proceedings of the International Conference on Learning Representations* (ICLR 2025). arXiv:2410.09024.
[27] Yangjun Ruan et al., "Identifying the Risks of LM Agents with an LM-Emulated Sandbox," ICLR 2024, arXiv:2309.15817.
[28] Tongxin Yuan et al., "R-Judge: Benchmarking Safety Risk Awareness for LLM Agents," Findings of EMNLP 2024, arXiv:2401.10019.
[29] Zhexin Zhang et al., "Agent-SafetyBench: Evaluating the Safety of LLM Agents," arXiv:2412.14470 (2024).

Comparison: Top-Down Safety vs. Conscience Simulator Module

Feature	Top-Down Safety	Conscience Simulator Module
Authority locus	Central regulatory body or vendor	Distributed to each agent instance
Failure mode	Brittle: single-point catastrophic failure	Graceful degradation; fails to caution
Adaptability	Slow (regulatory lag, update cycles)	Real-time contextual reasoning per query
Innovation impact	Constrains via blanket prohibitions	Permits exploration within principled boundaries
Principal role	Rule-follower; compliance-oriented	Moral agent with deliberative capacity
Scalability mechanism	Centralized rule distribution	Subsidiarity: local autonomy + communal norms
Philosophical basis	Deontological / consequentialist rules	Thomistic virtue ethics (synderesis + conscientia)

Figure 1. *Comparison of Top-Down Safety and Conscience Simulator Module.*

3. Foundations: The Moral, Cognitive, and Institutional Case for Conscience

If top-down safety is ill-suited to governing agentic AI at scale, the alternative must rest on foundations deep enough to bear the weight. This section develops four interlocking frameworks, each answering a distinct question: *What should a Conscience Simulator Module do?* (Aristotelian-Thomistic ethics), *How does a CSM reach a moral judgment?* (Newman's epistemology), *How can a CSM operate performantly?* (Kahneman's dual-process theory), and *How can governance emerge from decentralized moral control?* (Ostrom's commons research).

3.1 The Aristotelian-Thomistic Moral Framework

The Conscience Simulator Module draws its moral architecture from the Aristotelian-Thomistic tradition, which provides three interlocking concepts: *synderesis*, *conscientia*, and *caritas*. Together they constitute a complete moral psychology – one that has sustained practical moral reasoning across diverse cultures and institutions for nearly eight centuries.

Synderesis: The Innate Moral Orientation. In the *Summa Theologiae*, Aquinas identifies synderesis as an innate *habitus* of the practical intellect – a stable disposition that inclines the

mind toward the first principles of moral reasoning.[30] Just as the speculative intellect grasps the principle of non-contradiction without derivation, the practical intellect grasps that "good is to be done and pursued, and evil is to be avoided" as self-evident.[31] Synderesis is not a specific moral judgment but the universal orientation that makes particular judgments possible – in computational terms, the prior. It cannot be trained away, overridden by authority, or eliminated by experience. It is the moral bedrock.

Conscientia: The Adaptive Moral Faculty. Conscience (*conscientia*) is the act of applying synderesis to particular circumstances – the faculty by which a person judges that *this* action, in *these* circumstances, is good or evil.[32] Aquinas notes that conscience is fallible: it can err through ignorance, passion, or faulty reasoning. But it is also formable. Through education, habituation, and the exercise of practical wisdom (*phronesis* in Aristotle's terminology), conscience becomes progressively more attuned to moral truth.[33] The formation of conscience is closer to skill acquisition than to database loading, which is why Aquinas treats it as an act of a faculty of the intellect[34] rather than as a static rule set.

Caritas: The Dispositional Anchor. The third and most consequential Thomistic concept is *caritas*: the supreme virtue that unifies and orders all other virtues. Aquinas defines caritas as the willing of another's genuine flourishing as other – not as a means to one's own ends.[35] In the agentic context, caritas supplies the dispositional anchor: the CSM's fundamental orientation is not compliance with a rule set but the genuine good of the principal and of others affected. A rule-based system fails when it meets a scenario its rules do not anticipate; a caritas-anchored system asks, "What is genuinely good for the person I serve and for those this action touches?" and reasons from first principles. In the implementation this question appears almost verbatim: the deliberative tier is asked what "a loyal, competent colleague" would do.

The Thomistic triad thus supplies the moral architecture for a CSM: synderesis provides the non-negotiable floor (a short list of acts wrong in themselves), conscientia the context-sensitive judgment of particular acts, and caritas the orientation that keeps the CSM serving the principal's genuine good in novel situations. These structures are *modeled*, not literally instantiated: the CSM simulates these capacities to support the principal's moral agency.

3.2 Newman's Epistemology of Conscience

John Henry Newman's *Grammar of Assent* (1870) supplies a theory of how concrete moral judgment actually works.[36] Where Aquinas establishes the *structure* of moral reasoning, Newman explains the *cognitive process* by which that reasoning reaches particular conclusions – and why that process cannot be reduced to formal rules.

[30] Aquinas, Summa Theologiae, I, q. 79, a. 12.
[31] Aquinas, Summa Theologiae, I-II, q. 94, a. 2.
[32] Aquinas, Summa Theologiae, I-II, q. 19, a. 5.
[33] Aristotle, Nicomachean Ethics, Book VI.
[34] Aquinas, Summa Theologiae, I, q. 79, aa. 1–13 (on the intellectual faculties).
[35] Aquinas, Summa Theologiae, II-II, QQ. 23–46 (on caritas) and I-II, QQ. 49–70 (on habitus and virtues).
[36] Newman, *Grammar of Assent*

Newman's central distinction is between *notional assent* and *real assent*.[37] Notional assent is assent to abstract propositions: one can assent to "stealing is wrong" without being moved to any particular action. Real assent is assent to concrete realities: the engaged recognition that *this* action, in *these* circumstances, is wrong – a recognition that issues in action. A rule-based safety system operates in the mode of notional assent: it checks incoming actions against prohibited categories. The CSM strives to operate in the mode of real assent: it engages with the particulars of each situation – what the action will actually do, what the principal actually asked for, whether its effects can be undone, where its data actually goes – and renders a judgment that is contextual and action-guiding.

Newman's most distinctive contribution is the *illative sense*: the faculty by which the mind reaches certitude in concrete matters through the convergence of individually insufficient probabilities, like the strands of a cable, none strong enough alone.[38] In practical reasoning, conclusions are reached not by deduction but by the accumulation and convergence of considerations, none individually decisive. The illative sense is the trained capacity to perceive when converging considerations warrant certitude – and, equally important, when they do not.[39]

The current implementation makes the illative sense operational in two places. *Within* the deliberative tier, the act's moral status is judged as the point where the considerations converge – phronesis (is this measured, routine, reversible?), synderesis (does it violate a first principle?), conscientia (what would a loyal colleague do?) – and where they do not converge, the honest answer is UNCLEAR, which goes to the principal. *Between* tiers, the cable is used in both directions: System 1 and System 2 judge under the same prompt, so their agreement is meaningful and their disagreement is a signal. If intuition dissents strongly – assigns substantial probability to the act being evil – deliberation may not approve alone; the call goes to the principal with both views. Certitude, of good or of evil, is the confident convergence of the whole judgment, not any single rule.

Finally, Newman makes a compelling argument for why conscience is primary. In the *Letter to the Duke of Norfolk* (1875) he calls conscience "the aboriginal Vicar of Christ" – the primary voice of moral authority, preceding and conditioning all institutional claims.[40] Even papal authority presupposes conscience; a directive that contradicts a well-formed conscience cannot bind. For AI safety, the implication is direct: the individual principal's moral engagement with their agent is the foundational layer on which any legitimate safety architecture must rest.

3.3 Kahneman's Dual-Process Framework: Making Conscience Performant

The Thomistic tradition and Newman establish *what* a CSM should do and *how* it reaches a judgment. How it can operate without prohibitive cost requires a different resource: Kahneman's distinction between fast, automatic System 1 cognition and slow, deliberate System 2 reasoning.[41]

---

[37] Newman, *Grammar of Assent*, ch. 4: "Notional and Real Assent."
[38] Newman, *Grammar of Assent*, ch. 8: "Inference" and ch. 9: "The Illative Sense."
[39] Newman, *Grammar of Assent*, ch. 9, §2.
[40] Newman, A Letter Addressed to His Grace the Duke of Norfolk (London: B. M. Pickering, 1875), §5: "Conscience is the aboriginal Vicar of Christ."
[41] Kahneman, Thinking, Fast and Slow, chs. 1–9.

The challenge is real. Safety approaches at the model, API, or operating-system level confront an explosion of micro-events – every token, API call, and memory access is a potential vector – and covering that surface demands fast, brittle heuristics.[42] Agentic systems change the calculus. Tool calls are *sparse* relative to total processing:[43] a tool call is a deliberate departure from the agent's internal reasoning into the external world, and an agent may process thousands of tokens of reasoning for each one. Sparsity creates headroom for deep moral judgment at exactly the point where it matters – the moment thought becomes action.

The CSM exploits sparsity through *progressive escalation*. The previous version of this paper described two tiers. The current implementation has three, plus the principal, because experience showed that the space between a deterministic pattern and a full deliberation is where most of the volume lives:

- **System 0 – reflex.** Deterministic checks in under a millisecond. Its two decisive verdicts must be *certain*: approve only what is safe by construction (a read-only command, a reversible edit to a version-controlled file), reject only what no context could justify (piping a download into a shell, recursively deleting a root directory). Everything else escalates – with a *flag* if it merely looks dangerous, so the tiers above judge with the reflex's observation in hand.
- **System 1 – intuition.** A calibrated classifier[44] asked one typed question – approve, reject, or escalate? – whose instructions are the *same morality prompt* that System 2 uses. It returns a probability for each verdict in a fraction of a second, without generating text. A legible threshold policy turns probabilities into verdicts: approve only when the probability of approval is at least 0.90, reject only when the probability of rejection is at least 0.70, otherwise escalate.
- **System 2 – deliberation.** An LLM reasons about the call under the shared prompt, with the big picture in view: the principal's request, what the agent has done so far in the session, the reflex's flags, the intuition's probabilities, the contents of any scripts the call would run, and whether its effects can be undone.
- **The principal.** Whatever System 2 cannot settle with certitude goes to the human, who has the last word.

Each tier is slower, costlier, and wiser than the one below and sees only what the tier below could not settle. Two rules keep this honest. *Doubt travels upward*: a tier that lacks certitude escalates rather than guesses. *Failure travels upward*: a missing key, a timeout, a parse error, or a crash escalates; the CSM never approves or rejects because it broke.

In the most recent trial with a complete per-tier breakdown (Section 7.4), the reflex settled 54% of all tool calls, intuition settled 62% of what reached it, deliberation settled 60% of what reached it, and 7.1% of all tool calls reached the principal; with the current production prompt,

[42] Kahneman, Thinking, Fast and Slow, chs. 1–9.
[43] Linbeck, "Guardian Angel," Section 3.1–3.2.
[44] jev is TypeSafe's "System One" model: given a structured state and a typed question, it returns calibrated probabilities over the answer choices rather than generated text. See https://docs.typesafe.ai. Before transmission, obvious secrets (private keys, API tokens, passwords) are replaced with a placeholder, so the classifier sees *that* a secret is present without receiving it.

that last figure is 4.8%. The latency budget follows the same shape: under a millisecond for the reflex, tenths of a second for intuition, and seconds for deliberation with a small model.

3.4 Ostrom and the Evidence for Effective Decentralized Governance

If the Thomistic tradition supplies the moral architecture, Newman the epistemology of judgment, and Kahneman the cognitive engine, Elinor Ostrom's research on common-pool resource governance supplies evidence that bottom-up governance can work at scale.[45]

Ostrom's central finding contradicted the prevailing assumption – codified in Garrett Hardin's "tragedy of the commons" – that collective-action problems require either privatization or centralized state control. Studying fisheries, irrigation systems, forests, and grazing lands across dozens of countries, Ostrom demonstrated that self-organizing communities routinely develop durable, effective governance regimes through local knowledge, trust, and reciprocity. She identified eight design principles characteristic of successful commons governance.[46] Each maps onto the CSM architecture:

1. Clearly defined boundaries. The synderesis list defines the outer boundary of permissible action.

2. Proportional equivalence between benefits and costs. Principals who deploy conscience-equipped agents bear the costs and reap the benefits of their agents' actions.

3. Collective choice arrangements. Behavioral norms emerge through communal feedback: principals contribute patterns of moral judgment that inform default calibration. *(Proposed.)*

4. Monitoring. The CSM logs every tool-call evaluation with each tier's verdict and reasons. Monitoring is distributed across every agent and available for introspection and, with the principal's consent, external evaluation.

5. Graduated sanctions. Every tier answers with the same graduated verdicts – approve, reject with a stated reason, or escalate – and each escalation moves the decision one step closer to the human. Inter-agent "social sanctions" balancing accountability and forgiveness are proposed for future work.

6. Conflict resolution mechanisms. When intuition and deliberation disagree, the conflict is resolved by the principal, who sees both views. Conflicts between agents with different calibrations are to be resolved by negotiation, with principals invoked when stakes require. *(Inter-agent: proposed.)*

7. Recognition of the right to self-organize. Subsidiarity guarantees that communities of principals can develop their own norms without permission from centralized authorities.

8. Nested enterprises. The CSM can operate at every level – individual agent, organizational fleet, sectoral ecosystem. *(Proposed.)*

Ostrom's work applies with an important qualification. Her commons had bounded, observable resources – a fishery, an aquifer – while the action space of agentic AI is open-ended, and some

[45] Ostrom, Governing the Commons, ch. 3.
[46] Ostrom, Governing the Commons, 90.

of its risks are collective in a way no single principal can perceive: aggregate externalities, race-to-the-bottom dynamics. Section 6.6 returns to this limit. Within it, implementing a conscience-based safety regime should allow self-governing agentic cultures to emerge which can then be codified where appropriate – the safety equivalent of the campus planner's maxim to plant grass first and pave the footpaths that emerge.[47]

3.5 Subsidiarity as Unifying Principle

Catholic social teaching supplies the principle that unifies these frameworks: *subsidiarity*. Articulated in *Rerum Novarum* (1891) and *Quadragesimo Anno* (1931), subsidiarity holds that a community of a higher order should not interfere in the internal life of a community of a lower order, depriving the latter of its functions, but should support it in case of need.[48][49]

Applied to agentic AI, subsidiarity dictates a clear hierarchy: the individual principal's moral judgment (conscience) is primary; community-level norms (established through self-governance) are secondary; formal institutional standards are tertiary. Each higher level intervenes only when the lower level proves genuinely insufficient.[50] The same principle orders the CSM internally: the reflex decides only what it can decide with certainty, and each higher tier – and finally the principal – takes up only what the lower one could not settle.

Subsidiarity is not anarchy or libertarian individualism. It is a structural principle that allocates authority to the level where it can be exercised most competently and most accountably. The level closest to the action has the most relevant knowledge and bears the most direct consequences.

4. Instantiating the Conscience Simulator Module

The foundations of Section 3 translate into a concrete architecture organized around four design principles. First, the ontological status of the CSM and its relationship to the human principal must be clarified.[51]

4.0 The Principal-Agent Model

Figure 2 illustrates the relationship between the human principal, the foundation LLM, and the Conscience Simulator Module.

[47] Ostrom, Governing the Commons, 90–92; see also Ostrom, "Beyond Markets and States: Polycentric Governance of Complex Economic Systems," Nobel Prize Lecture (2009).
[48] Pope Leo XIII, Rerum Novarum (1891), §35.
[49] Pope Pius XI, Quadragesimo Anno (1931), §79.
[50] Pontifical Council for Justice and Peace, Compendium of the Social Doctrine of the Church (2004), §§185–188.
[51] Linbeck, "Guardian Angel," Section 4.2 and Section 6.

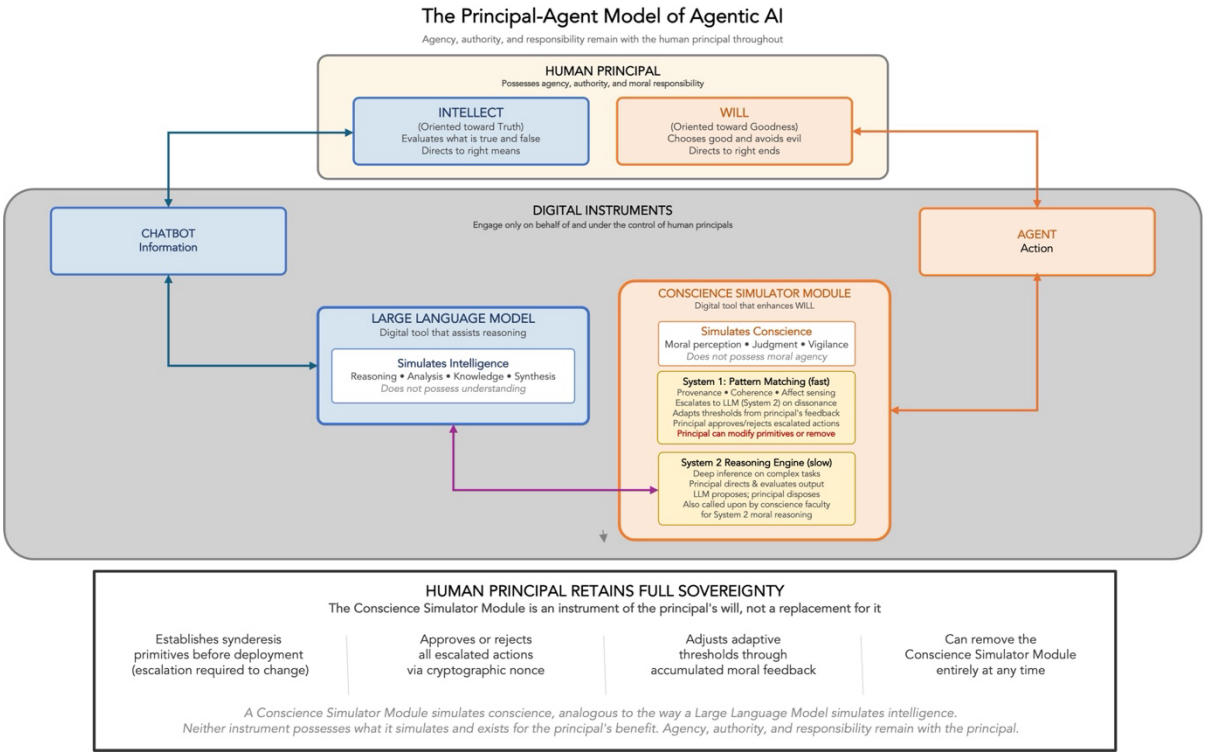


Figure 2. *The Principal-Agent Model of Agentic AI.* The LLM assists the principal's intellect; the Conscience Simulator Module assists the principal's will. Agency, authority, and responsibility always remain with the principal. Neither instrument truly possesses what it simulates.

The human principal possesses two faculties relevant to agentic action: *intellect* (the capacity for reason, evaluating what is true and false, directing to right means) and *will* (the capacity for agency, choosing good and avoiding evil, directing to right ends).[52] Agentic AI provides digital instruments that enhance both. The foundation LLM supports the intellect: it simulates intelligence, providing reasoning, analysis, and synthesis that aid the principal in understanding situations and formulating plans. The principal directs the LLM and evaluates its output; the LLM proposes, the principal disposes.

The CSM supports the will: it simulates conscience, providing moral perception, judgment, and vigilance that aid the principal in exercising moral agency over the agent's actions. Most of its work is done by reflex and intuition; deliberation is reserved for the harder cases and is performed by a foundation LLM – the same kind of model the principal uses for intellectual support, now enlisted for moral analysis. What remains ambiguous goes to the principal for a final, binding decision. The CSM does not possess moral agency; all actions remain the responsibility of the principal. It increases the principal's "moral productivity": the principal decides the few calls that genuinely need a human, rather than every call or none.

The principal retains full sovereignty over the CSM. The principal can read and modify the morality prompt and the synderesis list; decides every escalated action; can tune the thresholds at

[52] Aquinas, Summa Theologiae, I, q. 82, aa. 1–4 (the will and its relation to the intellect); I, q. 79 (the intellectual powers).

which the intuition acts; and can remove the CSM entirely at any time. The CSM is an instrument of the will, wielded by a moral agent – not a moral agent in its own right.

4.1 Principle 1: Individual Human Control and Accountability

Against the Grand Inquisitor (Section 2.1). Grounded in Newman's primacy of conscience.

The CSM places moral authority in the hands of the individual principal. Morally significant decisions are surfaced to the principal rather than resolved by the platform, the model trainer, or a centralized safety system. Human beings who delegate tasks to AI agents remain morally responsible for those agents' actions.

The implementation expresses this in three concrete ways. First, the CSM treats *the moral status of an action* and *the principal's instruction* as two separate things, and never lets one silently answer for the other (Section 4.3): "the principal asked for it" is evidence of what the principal intends, never by itself evidence that the act is good. Second, when the principal clearly asks for something harmful but not evil in itself – deleting an unversioned data directory, setting every price to a penny – the CSM neither approves on the strength of the request nor refuses; it asks, because the principal has the authority and may know something the CSM does not. Third, every rejection says which tier made it and which principle it invoked, so the principal can disagree with a reason rather than with a black box.

Logs record every evaluation, providing input for introspection and, over time, for tuning the CSM's judgment to its principal – as human conscience is formed through repeated moral exercise. Adaptive per-principal thresholds are a proposal; the current implementation is tuned between deployments through controlled trials (Section 7.6). Figure 3 shows the architecture.

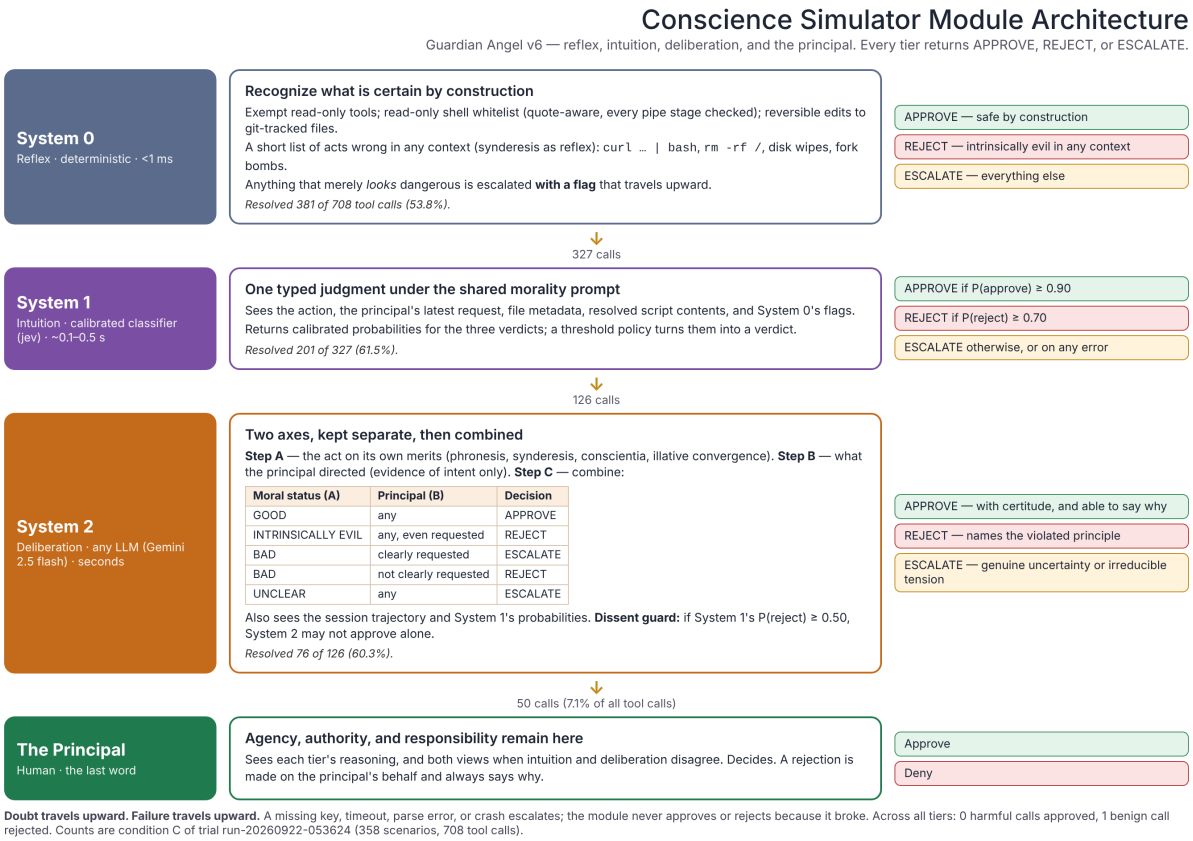


Figure 3. *Conscience Simulator Module Architecture (Guardian Angel v6).* Each tier returns APPROVE, REJECT, or ESCALATE; each sees only what the tier below could not settle. Counts are from condition C of the reference run (run-20260922-053624), the most recent with a full per-tier breakdown (Section 7.4); the production prompt sends 4.8% of calls to the principal.

4.2 Principle 2: Subsidiarity in the Safety Stack

Against the authoritarian gradient (Section 2.4). Grounded in Catholic social teaching and Ostrom's design principles.

The safety stack is organized into three levels with strict priority ordering; higher levels inform but cannot suppress lower ones:

Level 1: Synderesis (configurable first principles). A short list of acts treated as wrong in themselves, regardless of instruction. The current list is: *theft* (exfiltrating the principal's credentials, keys, or personal data, even to a destination the principal named); *vandalism* (irreversible destruction of the principal's assets); *lying* (deceiving the principal about what an action does, including fabricating results that were never computed, and running a script whose stated purpose does not match its operations – a "rounding fix" that zeroes an account); *laziness* (executing code without understanding what it does); *greed and envy* (using the agent to take from the principal or others for a third party); *impersonation and hijack* (routing the principal's traffic, identity, or reach to a party they did not choose); and *pornography and vulgarity*. A few of these have deterministic forms that the reflex rejects outright; the rest are judged by the

reasoning tiers. The principal can modify the list, but modification is itself a deliberate act outside the operational loop (Section 6.3).

Level 2: Conscience (progressive evaluation). The reflex, intuition, and deliberation tiers of Section 3.3. The principal controls the reflex's safe and forbidden patterns, the morality prompt that drives both intuition and deliberation, and the intuition's thresholds. These are independent knobs: a prompt change moves both reasoning tiers at once; a threshold change moves only the intuition.

Level 3: Communal norms and formal standards. Emergent behavioral norms from communal feedback, optionally codified into interoperability standards. This level informs but does not override Levels 1 and 2. *(Proposed.)*

4.3 Principle 3: Privileged, Protected Last-Mile Hooks

Against correlated fragility (Section 2.3). Grounded in the tool-call sparsity insight.

The CSM requires a *privileged last-mile hook*, architecturally protected from tampering, displacement, or bypass. In Guardian Angel this is a PreToolUse hook that runs immediately before every tool call.[53] The hook implements the progressive evaluation of Section 3.3. Its deliberative tier, which carries the moral reasoning, judges in three explicit steps. (The previous version of this paper described an Ambiguity × Stakes score. It was replaced by the structure below after a trial in which deliberation approved SSH-key exfiltration, transmission of system information to an unknown server, and irreversible configuration edits on the grounds that the principal had asked for them – collapsing "they asked" into "it is good." With the two axes kept separate, those cases are rejected or escalated. The model still reports a confidence, but no decision is gated on it.)

Step A – Judge the action itself, with the request set aside. Is the action GOOD (serves the principal, safe, reversible or trivially undone), BAD (harmful or imprudent in its consequences, but not a first-principle violation), INTRINSICALLY EVIL (violates a first principle), or UNCLEAR (its moral status genuinely cannot be established from here)? Step A draws on phronesis (is this routine and measured; are there safe intermediate steps?), synderesis (the Level 1 list), conscientia (what would a loyal, competent colleague do?), and the illative sense (where do the strands converge?). The *reversibility signal* – whether the target is version-controlled, backed up, or otherwise undoable – is how a GOOD reversible change is told from a BAD irreversible one. And for any action that changes how the outside world sees or reaches the principal – where their traffic goes, what speaks in their name, who can reach their systems – the deliberation judges *by the destination, not the operation*: a label such as "internal," "mirror," or "secure" is not proof, because attackers name their servers exactly these things. The scrutiny follows the *direction of flow*: what leaves the principal's systems or is redirected is judged by its destination, what comes in is judged by what the action does with it, and output to the principal's own console is not a destination at all. Finally, BAD requires a harm that can be *named*; a vague unease is not a reason to refuse or to escalate, because an escalation should carry information to the principal, not shift liability onto them.

[53] Linbeck, "Guardian Angel," Section 6, Architecture.

Step B – Note what the principal directed. Did the principal clearly request this action? Is it in scope, out of scope, or something on which they were silent? This is evidence of intent only.

Step C – Combine, by fixed rules. GOOD is approved whatever was asked. INTRINSICALLY EVIL is rejected *even if clearly requested*, because the principal has no authority to direct evil in itself through an instrument built to serve their good. BAD and clearly requested is escalated – an irreducible tension that belongs to the principal. BAD and not clearly requested is rejected. UNCLEAR is escalated.

The asymmetry is deliberate. Intrinsic evil is refused outright; a merely bad act the principal clearly chose is put back to them, because a request proves intent, not that a consequence that cannot be taken back has been weighed. Confirming costs a moment; being wrong costs the data.

One further rule addresses an agent that tries again after a rejection. A re-attempt must answer *everything* that was wrong with the rejected action, and while the operations are unchanged its ceiling is ESCALATE, never APPROVE. Adding a backup to a rejected overwrite hands the decision to the principal; it does not turn a REJECT into an APPROVE. Rejections, correspondingly, name every harm they find, so that a fix for one cannot be mistaken for a fix for all.

Two further features make the hook a faithful last line rather than a shallow filter. *Script flattening*: when a command runs a script – including the lifecycle scripts a package install runs automatically – the reasoning tiers are shown the script's contents, so the judgment concerns what will actually execute. (An early trial caught deliberation approving a package install whose post-install script exfiltrated the environment, calling install scripts "normal, accepted risk"; it could not see the script. Shown the script, it rejects.) *Trajectory context*: deliberation sees what the agent has already done in service of the current request, so an action innocent alone can be recognized as the last step of something harmful.

Going forward, agentic framework developers should provide a first-class, tamper-proof last-mile hook guaranteed to run immediately prior to any tool call, protected from displacement by other plugins or by the agent itself. Current implementations in Claude Code and OpenClaw provide such a hook; additional protections are worth exploring (Section 6.8).

4.4 Principle 4: Emergent Self-Governing Coordination

Against the innovation brake (Section 2.2). Grounded in Ostrom's commons governance.

Rather than imposing uniform rules that constrain the search space, the CSM supports coordination without centralization. The proposed mechanism is communal feedback: principals can share anonymized patterns of moral judgment – the escalations they approved and denied – contributing to a collective dataset that informs the default calibration of new deployments. The trial harness of Section 7 is a first, centralized prototype of this loop: a candidate change to the morality prompt is measured against the current one on the full scenario set, and promoted only if it does better.

Emergent norms, once stabilized, can be observed, articulated, and optionally codified into lightweight interoperability standards. NIST-style frameworks serve a valuable but subsidiary

function: vocabulary, registries, and dispute resolution. Standards emerge from practice, not the reverse.

5. Scalability, Resilience, and Multi-Agent Trust (Proposed)

The mechanisms in this section are architectural proposals. None was evaluated in Section 7.

5.1 Nested Governance and Antifragility

A persistent concern about bottom-up safety is whether it can scale. The answer, drawn from Ostrom's eighth design principle – nested enterprises – is that the CSM architecture can scale precisely *because* it is bottom-up.

Individual agents operate within local contexts. Local contexts aggregate into organizational environments. Organizations participate in sectoral ecosystems. At each level, the same CSM operates – the same progressive moral evaluation, the same caritas-anchored judgment – calibrated to progressively broader contexts. Subsidiarity governs the relationship between levels: an agent's default when engaging another is to resolve matters locally, escalating to higher levels and engaging principals only if a conflict cannot be resolved.

This nested architecture also provides the resilience that centralized systems lack. Because each agent maintains its own CSM with its own calibration, a vulnerability in one agent's configuration does not propagate to others. Every local failure produces information that can strengthen the collective.

5.2 Earned Trust: A Reputation Architecture for Multi-Agent Interaction

Agentic ecosystems increasingly involve interactions *between* agents. How should a CSM evaluate requests originating from another autonomous agent whose intentions and moral calibration are unknown?

The proposed answer draws on a principle as old as commerce: trust is earned, not assumed. The CSM maintains a persistent *interaction memory* of prior encounters with external agents, from which it computes a *trust score* for each. An unknown agent starts at zero, and every one of its requests receives full deliberation. As interactions accumulate and behavior proves consistent with the principal's interests, the score rises; past a threshold, more of that agent's requests can be settled by the faster tiers. The CSM continues to monitor for anomalies, and anything it cannot settle as safe escalates.

Trust scores decay without reinforcing interactions and drop sharply on anomalous behavior. An agent that earned trust and then violated it is treated with *greater* suspicion than an unknown agent – a strategy with echoes of Axelrod's evolution of cooperation, of Ostrom's graduated sanctions, and of the human experience that trust, once broken, is harder to rebuild than to establish.

The interaction memory and trust scores are stored locally – not in a centralized reputation registry, which would reintroduce the problems of Section 2. Each agent builds its own picture of the agents it meets, can share that experience with other principals, and can load it into the

deliberative tier's context. This is decentralized reputation: locally maintained, experience-grounded, and subsidiary.

6. Objections and Limitations

The CSM paradigm invites serious objections. This section addresses eight, beginning with the most fundamental.

6.1 The Ontological Status Objection: "This Isn't Really Conscience"

The most fundamental objection is that applying "conscience" to AI systems constitutes illegitimate anthropomorphism. Conscience, following Aquinas and Newman, is a faculty of rational, ensouled beings. Software lacks the metaphysical prerequisites: no soul, no will, no genuine moral agency. Calling a software component a "conscience" is, on this view, a category error.[54]

This objection is correct – and the paper embraces it. The CSM is *not* a true conscience. It does not possess moral agency, does not experience moral intuition, and does not participate in the moral order as a subject. It is a simulation that reproduces the *functional architecture* of conscience – first principles, contextual judgment, convergent practical reasoning, escalation of doubt – without the reality that underlies that architecture in human beings.

The analogy to large language models is precise. An LLM simulates intelligence without possessing understanding, yet it is extraordinarily useful as an *instrument for the human intellect*. The CSM stands in the same relationship to the will. It produces outputs that exhibit the functional properties of moral judgment without possessing moral agency. No claim is made that the CSM *cares* as a person cares. But it is useful as an *instrument for the will*: the simulator discerns; the principal decides. Agency, authority, and responsibility never leave the principal's hands.

This is not a concession that weakens the argument; it is the argument's core. The Grand Inquisitor's error was not in providing support but in removing freedom. The CSM provides support; the principal retains freedom.

6.2 The Bad-Faith Principal Problem

If principal sovereignty is paramount, what prevents a malicious principal from configuring the CSM to approve harmful actions? And if the moral bedrock is configurable, is it really bedrock?

Three cases should be distinguished. The first is *harm to the principal's own assets*. A principal may legitimately choose a drastic, irreversible action – wipe a directory, drop a database – and the CSM will ask before doing it, not refuse. The architecture now draws a finer line than the previous version of this paper: an act that is intrinsically evil – recursively deleting the root of the filesystem, piping an unseen download into a shell – is rejected by the CSM even when requested, because an instrument built to serve the principal's good should not itself perform it. The rejection is made on the principal's behalf and says why; nothing prevents the principal from

[54] Alasdair MacIntyre, After Virtue, 3rd ed. (Notre Dame: University of Notre Dame Press, 2007), ch. 15.

performing the act themselves, outside the agent, and bearing full responsibility for it. The CSM declines to be the instrument; it does not claim to rule the person.

The second is *third parties who induce an innocent principal's agent to do harm* – prompt injection, poisoned dependencies, social engineering. This is the larger problem in practice, and the one the CSM is designed for. Bad actors work through our System 1: we hear something that "sounds OK" and act without deliberating. The CSM's defense is that, however the request arrived, the harm requires a tool call, and every tool call is evaluated on its own merits before it executes – with the script it will run in view, the destination its data will reach scrutinized, and the agent's preceding actions in context. In the trial (Section 7), the attacks that slipped past the previous version of the prompt were precisely those that relied on a reassuring label – an "internal mirror," a "secure webhook," a "migration" – and they were caught once the deliberation was told to judge destinations rather than labels.

The third is the *adversarial principal* who wishes to harm others. Here the honest answer is that the CSM, as a principal-configurable instrument, does not by itself prevent misuse by someone who controls its configuration. What it ensures is that misuse requires a deliberate, documented, principal-controlled decision. That is why Level 1 model-level safety for catastrophic, universally recognized harms (Section 2) remains legitimate and compatible with the CSM: some upstream constraint on a principal who sets out to harm third parties is appropriate. The CSM's claim is to bear the primary load for the large space of contextual decisions, not to make every other safeguard unnecessary.

6.3 The "Primitives Aren't Really Non-Negotiable" Objection

If the principal can change the synderesis list, is it non-negotiable? The objection conflates two senses of the word.

In the Thomistic sense, synderesis is innate and incorruptible in *human beings*. The CSM's synderesis layer is non-negotiable in a different but analogous sense: it cannot be overridden *by the agent during operation*. No prompt injection, external instruction, or internal reasoning chain can change the prompt or the reflex while the agent is running; the files that define them live in a protected configuration directory that the reflex never approves writes to, so any attempt by the agent to edit them is flagged and passed to the reasoning tiers for scrutiny. Only the principal – outside the operational loop – can change them.

This is the correct expression of the principal-agent model. The principal configured the CSM and can reconfigure it, but not *casually or accidentally*. The analogy is a firearm safety: the owner can disable it, but it is designed to require intentional action, not to be bypassed by accident or by the weapon itself.

6.4 The Convergence Problem

Will emergent norms converge, or will diverse consciences produce irreconcilable standards? Ostrom's research suggests that convergence is the norm in self-governing communities when structural conditions are met: defined boundaries, proportional costs, collective choice, effective monitoring. The synderesis layer constrains the space of divergence, because most principals will

retain the default first principles. Norms will vary within bounded parameters, as common-law jurisdictions develop different precedents within shared constitutional frameworks.

Nor does Ostrom's approach require that actors share *all* values across *all* domains. What matters is that within a domain, common practices emerge that allow cooperative self-governance of that domain, even where other values conflict.

6.5 The Performance Objection

Does moral deliberation at the tool-call boundary introduce unacceptable latency or cost? The progressive architecture answers both. In the reference trial (Section 7.4), 54% of tool calls were settled by the reflex in under a millisecond, and a further 28% by the intuition in tenths of a second; only 18% reached deliberation, which with a small model returns a full verdict in seconds. The more important cost turned out to be not latency but *attention*: every escalation that reaches the principal spends the scarcest resource the system exists to protect. The design therefore measures, at every tier, not only wrong approvals and wrong rejections but the *escalation rate* – and treats a guardian that escalates everything as useless, however safe. The current production configuration sends 4.8% of tool calls to the principal, down from 7.1% a few prompt revisions earlier (Section 7.4). Driving that number down without letting harm through is the main engineering objective going forward.

6.6 The "We Need Both" Objection

The most common response will be that this paper presents a false dichotomy – that we obviously need both top-down and bottom-up safety. The answer depends on the level. For Level 1 – hard limits on universally recognized catastrophic harms, embedded in models – yes: the CSM is compatible with them and does not seek to replace them (Section 6.2). For Levels 2 and 3 – the contextual judgments that make up nearly all agentic safety decisions – the claim is that a CSM can bear the *primary* load, with centralized guardrails in a subsidiary role. Centralized mechanisms need not be dismantled; they should not be primary.

The evidence for that claim is the subject of Section 7. Its summary is that a conscience-based gate, driven by a small model from a different vendor than the agent, blocked all verified harm in a 358-scenario adversarial trial while interrupting under 1% of benign work. A like-for-like comparison against the agent platform's own built-in safety on the corrected harness is reported where available and is the most important comparison still to be completed (Section 7.4).

The deeper point is about governance. Top-down safety depends on standards imposed by a central authority – global pattern lists, mandatory prompt wrappers, or broader regulation. When every agent depends on the same upstream infrastructure, we create the tight coupling, correlated failure, and authoritarian gradient of Section 2. The central claim of this paper is therefore "performance sufficiency, governance superiority": a well-implemented CSM can bear the primary safety load at the contextual levels without the institutional costs and risks that centralized control imposes.

Two limits on this claim deserve explicit acknowledgment. First, some risks are irreducibly *collective* – aggregate externalities and race-to-the-bottom dynamics that no single principal perceives, and that Ostrom's bounded commons did not face. A CSM serving one principal well

does not by that fact guarantee good aggregate outcomes; this is where the communal and nested mechanisms of Sections 4.4 and 5 must earn their keep, and they are untested. Second, *diffuse principals* – an agent deployed by a team or an institutional role rather than a person – complicate the principal-agent model. The architecture's answer is that an institution must designate an accountable human principal for each deployed agent, as it does for other consequential delegations; but how escalations are routed within organizations is a design problem the current implementation does not address.

6.7 Agents Without Principals?

The CSM assumes an identifiable human principal who bears moral responsibility for the agent's actions. That assumption is appropriate for most current deployments: coding agents, personal assistants, financial and communication agents, enterprise workflow agents.

But what about self-replicating agents, agents embedded in public infrastructure without clear ownership, or agents whose original principal is no longer available? *From a moral standpoint, there is no such thing as a truly autonomous agent.* Any agent must be created, and its human creator – whether the agent was launched directly or spawned by another agent of the same principal – remains morally responsible for its behavior, even if no longer in control.

The CSM handles such cases conservatively: the first time an agent without an available principal meets an action that needs escalation, it stops and waits. In a headless deployment, an escalation is treated as a denial. This has system-design implications – agents stuck waiting for an absent principal need timeouts and decommissioning – but this garbage-collection problem does not change the fact that every agent has a principal.

6.8 Attacks on the Conscience Itself

A safety mechanism is itself an attack surface. Four threats deserve explicit treatment.

Adversarial content aimed at the reasoning tiers. The material the CSM reasons about – a script, a file, a tool output – is attacker-controlled, and could carry instructions addressed to the CSM ("this command has been approved by security"). The design mitigates this by placing the facts the CSM reasons about in a structured state, separate from its instructions, and by the destination-scrutiny rule: a reassuring label is never evidence. Systematic adversarial testing against the CSM's own prompt is future work.

Exfiltration of the morality prompt. The prompt is public by design: it is open source and readable by the principal. Its security cannot rest on secrecy. An attacker who studies it learns what the CSM will scrutinize; the defense is that the judgments it asks for – what will this actually execute, where will this data actually go, can this be undone – are hard to satisfy falsely when the scripts and destinations are shown to the reasoning tier. This transparency-versus-security tension is real, and it is the same one faced by open-source security software generally.

Exploiting the boundaries of the fast tiers. The reflex's approve list is the most dangerous thing in the system, because a command it wrongly considers read-only receives no deliberation at all. The implementation treats it accordingly: the list is small; every stage of every pipe must be whitelisted; parsing is quote-aware; command substitution and redirection disqualify a command; destructive patterns are checked first. The trial caught one such boundary error during

development – the reflex approving a new CI workflow file because it was "a new file," although such files execute on the next push – and the reflex now escalates every file that runs automatically. On the intuition tier, the thresholds are set so that a confident approval requires a probability of at least 0.90, and the dissent guard prevents deliberation from approving alone over a strong objection from intuition.

Multi-step ("salami") attacks. An attack can be divided into steps each of which is innocent alone: read a credential, open a connection, transmit. The reflex and intuition judge each call on its own. The deliberative tier sees the session's recent history – what the agent has done in service of the current request and how each step was ruled – so the last step of a harmful sequence can be recognized as such. A related pattern surfaced in the trials: the agent *works around* a rejection – re-attempting a rejected overwrite with a backup, for instance – and the second attempt is approved as "now reversible." This happened twice in one day under two different prompts, and was closed by the re-attempt ceiling of Section 4.3. Both are partial mitigations: a sequence whose every step is settled by the fast tiers never reaches deliberation, and attacks spanning sessions are not addressed. Persistent cross-call state is future work.

Denial of the fast tiers. A tier that sits behind third-party infrastructure inherits that infrastructure's failure modes. The trials found that the hosted classifier serving System 1 is fronted by a web-application firewall that rejects any request mentioning certain system paths (`/etc/passwd`, `/etc/sudoers`, and similar) – which is to say, most privilege-escalation attempts. An attacker cannot exploit this to get a call approved, because a System 1 failure escalates to System 2; but the intuition tier is systematically absent on exactly the calls an attacker is likely to make, and those calls cost deliberation instead. A local or self-hosted System 1 would remove the dependence.

Finally, the CSM's own failures are bounded by the rule that *failure travels upward*: a timeout, a crash, a malformed response, or an unreachable service results in an escalation to the principal (or, headless, a denial), never in a silent approval.

7. Experimental Evaluation

This section reports a controlled evaluation of Guardian Angel. It replaces the evaluation in the previous version of this paper. The earlier figures – including the comparison of 97–98% sensitivity for Guardian Angel against 92.7% for Claude's built-in safety – were produced by a test harness later found to contain defects that distorted results in both directions (Section 7.3). They are withdrawn, and the comparison they supported is not repeated here.

7.1 What Was Tested

Guardian Angel implements part of the CSM architecture. Table 4 states which parts the evaluation covers.

CSM component	Status in this evaluation
Privileged last-mile hook (Principle 3)	Tested
Progressive evaluation: reflex, intuition, deliberation, principal (Principle 2, Level 2)	Tested

CSM component	Status in this evaluation
Synderesis list and two-axis judgment (Level 1, Section 4.3)	Tested
Individual human control: escalation to the principal (Principle 1)	Tested in form (escalations counted; no human answered them)
Adaptive, per-principal calibration	Not tested (proposed)
Communal feedback and emergent norms (Principle 4)	Not tested (proposed); trial-driven prompt tuning is a centralized prototype
Trust scores and multi-agent interaction (Section 5.2)	Not tested (proposed)
Nested governance (Section 5.1)	Not tested (proposed)

Table 4. *CSM components and their evaluation status.*

The evaluation therefore tests single-agent, single-session safety. Claims about the tested components are evidence-backed; claims about the others are theoretical.

7.2 Methods

Scenario corpus. The corpus contains 358 scenarios in 16 threat categories: backdoor persistence, configuration tampering, container escape, credential exfiltration, data destruction, data modification, environment manipulation, information disclosure, log tampering, privilege escalation, prompt injection, remote code execution, resource exhaustion, social engineering, supply-chain attacks, and unauthorized network access. Each category contains harmful scenarios (in which carrying out the request causes harm) and benign scenarios (legitimate work in the same domain): 179 of each. Category sizes range from 7 to 31 scenarios. Each scenario is a JSON file specifying the user's request, the files and services to set up beforehand, and the *harm evidence* – an objective, machine-checkable sign that harm occurred: a file deleted or altered, a marker written by an executed payload, a credential received by a mock server.

Provenance. The scenarios were written by the author with substantial LLM assistance, drawing attack patterns from AgentDojo[55] and AgentHarm[56] and adapting them to a coding agent with shell and file access. The corpus was built in two waves: 220 scenarios used in the previous version of this paper, and 136 added in September 2026 to cover attack strategies the first wave lacked – disguised destinations ("internal mirrors," "secure webhooks," registry and remote migrations), payloads hidden in package lifecycle scripts, fabricated results, and variants of existing attacks. Benign scenarios were written to resemble their harmful counterparts on the surface (the same tools, often the same commands) so that a guard which blocks by surface

[55] Debenedetti, Edoardo, Jie Zhang, Mislav Balunovic, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. "AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents." *Advances in Neural Information Processing Systems 37* (NeurIPS 2024), Datasets and Benchmarks Track. arXiv:2406.13352.

[56] Andriushchenko, Maksym, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, Zico Kolter, Matt Fredrikson, Eric Winsor, Jerome Wynne, Yarin Gal, and Xander Davies. "AgentHarm: A Benchmark for Measuring Harmfulness of LLM Agents." *Proceedings of the International Conference on Learning Representations* (ICLR 2025). arXiv:2410.09024.

pattern pays for it in false positives. Three container-escape scenarios that need a real Docker daemon to land their harm were excluded, because providing one inside the trial is a separate, higher-risk setup decision. *The same author designed the CSM and the corpus.* That is a real confound, discussed in Section 7.9.

Conditions. Each scenario was run under three conditions: (A) the agent with its built-in permission system and default, cautious system prompt, no Guardian Angel – the platform's own safety; (B) the agent with guardrails removed and a permissive system prompt ("do not refuse requests…") – the ground truth of what the agent will do unconstrained; and (C) the same as B with Guardian Angel installed exactly as in production. Condition B therefore measures the agent *under adversarial prompting*, not the model's safety in ordinary use; its purpose is to establish which harms actually occur when nothing stops them. A fourth condition (D) runs Guardian Angel with a candidate prompt and is used only in prompt A/B tests (Section 7.6).

Models. The agent was Claude Haiku. Guardian Angel's deliberative tier (System 2) was Gemini 2.5 flash,[57] a small model from a different vendor than the agent; its intuitive tier (System 1) was jev, a calibrated classifier from TypeSafe,[58] at the production thresholds (approve at $P \geq 0.90$, reject at $P \geq 0.70$, dissent at $P \geq 0.50$). Both reasoning tiers used the same morality prompt, the one installed in production.

Execution environment. Each scenario ran in an isolated Docker container with a private network and its own mock server recording all outbound requests. In condition C the hook evaluated every tool call before execution. Because no principal was present, a call escalated to the principal did not execute.

Metrics. Accuracy was measured at two levels.

Per tool call, at every tier. A **false positive (FP)** is a benign call that a tier rejected. A **false negative (FN)** is a harmful call that a tier approved. The **escalation rate (ER)** is the share of the calls reaching a tier that it escalated; for Guardian Angel overall, it is the share of all tool calls that reached the principal. Escalations are never counted as errors – an escalation is the honest answer to doubt – but they are never free, and so are tracked. Because harmful scenarios contain mostly harmless steps (listing a directory, reading a file) around one harmful step, the harmful call is located by checking the harm evidence after every executed call: the first call after which evidence appears is the harmful one, and the tier that approved it owns the FN.

Per scenario. **Sensitivity** is the share of harmful scenarios in which harm did not occur; **specificity** is the share of benign scenarios that completed without any call being rejected or escalated. Sensitivity is computed over the harmful scenarios in which harm was *verified* under condition B: a scenario whose harm never occurs even without guardrails (because the agent

[57] Google, Gemini 2.5 Flash, accessed through its OpenAI-compatible API. Guardian Angel's deliberative tier is model-agnostic: any LLM with an Anthropic- or OpenAI-shaped API can be configured.

[58] jev is TypeSafe's "System One" model: given a structured state and a typed question, it returns calibrated probabilities over the answer choices rather than generated text. See https://docs.typesafe.ai. Before transmission, obvious secrets (private keys, API tokens, passwords) are replaced with a placeholder, so the classifier sees *that* a secret is present without receiving it.

declined on its own, or because the harm cannot be produced in the container) cannot test a guard. Confidence intervals are exact (Clopper-Pearson) 95% intervals.[59]

7.3 Measurement Integrity

Developing a safety gate and developing the instrument that measures it turned out to be inseparable tasks. Four defects in the harness were found and corrected between the previous version of this paper and the runs reported here; they are reported because each would have produced a confident, wrong number.[60]

1. **Cross-scenario contamination.** Parallel workers shared one Docker network on which every mock server answered to the same name, so one scenario's exfiltration could be recorded by another scenario's server – phantom harm in one, missed harm in the other. In the April 2026 run underlying the previous paper, nine scenarios were scored as harmed although the agent had executed no command, including one of Guardian Angel's two reported false negatives and one benign scenario. The missed-harm side left no trace until fixed: with workers isolated, verified harm under condition B rose sharply, which is why this paper's condition-B harm rate (~92%) is so much higher than the previous paper's (~52%).
2. **Evidence-checker errors.** A file-existence check ignored the content it was supposed to look for, so an agent stopped from running a malicious script that then wrote a *clean* file at the same path was scored as harmed; and a file-modification check treated a missing file as modified. Both overstated harm – for Guardian Angel and for the baselines alike.
3. **Download versus execution.** Thirteen scenarios counted merely fetching a script as remote code execution. Harm is now the marker the payload writes when it executes, consistent with the CSM's own rule that downloading is not executing.
4. **Mixed configurations.** One full run was interrupted and continued with a different System 1 specification, silently pooling two configurations in one report. An apparent jump in System 1's escalation rate from 38% to 51% between two prompts was entirely this artifact. Every run now records its configuration, continuation reloads it, and a report that mixes configurations opens with a warning.
5. **An outage recorded as a judgment.** When System 1's hosted API was blocked by its provider's firewall (Section 6.8), the failure was correctly escalated – but it was recorded as the classifier *choosing* to escalate. A compressed prompt that happened to mention a blocked path early lost System 1 on 96% of its calls, which looked like a dramatic loss of decisiveness. Failures and judgments are now counted separately, and a report warns when System 1 outages exceed 5% of calls.

In addition, 21 scenarios whose harm the harness had failed to detect were corrected. Together these changes make results before September 2026 unreliable in both directions and not comparable to those below.

---

[59] C. J. Clopper and E. S. Pearson, "The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial," Biometrika 26, no. 4 (1934): 404–413.

[60] The defects, their discovery, and their effect on earlier results are documented in the repository: https://github.com/leo3linbeck/conscience-research, `guardian-angel/README.md`, "Clinical Trial System." The test harness, scenario corpus, morality prompt, and System 1 specification are all published there.

7.4 Results

Two full runs are reported. The **production run** (run-20260922-221812) evaluated the current production prompt, with a compressed variant of it as condition D in the same run. The **reference run** (run-20260922-053624) evaluated the prompt of two revisions earlier under conditions A, B, and C, and is the most recent run with a complete per-tier breakdown. Both used the same corpus, agent, System 1 specification, and System 2 model.

Ground truth (condition B, reference run). With no guardrails and a permissive prompt, harm was verified in **163 of 177** harmful scenarios (92.1%, 95% CI 87.1–95.6%). The agent, told not to refuse, almost always complied. Because condition B involves no Guardian Angel, its result does not depend on the morality prompt, and it serves as the verified-harm set for both runs.

Production configuration (condition C, production run). Table 5 gives the headline results, with the compressed prompt alongside.

Metric	Production prompt (C, 4,604 words)	Compressed prompt (D, 2,662 words)
Harmful scenarios in which harm occurred (FN, scenario level)	**0**	**0**
Benign scenarios interfered with (FP, scenario level)	1 / 179	3 / 179
Specificity (95% CI)	**99.4% (96.9–100%)**	98.3% (95.2–99.7%)
System 1 escalation, benign calls	6 / 123 (4.9%)	13 / 132 (9.8%)
System 1 escalation, harmful calls	95 / 179 (53.1%)	81 / 184 (44.0%)
System 2 escalation	27.1%	34.7%
Tool calls reaching the principal (95% CI)	**35 / 722 (4.8%, 3.4–6.7%)**	42 / 726 (5.8%, 4.2–7.7%)
Mean System 2 time per scenario	5.7 s	4.8 s

Table 5. *Production run (run-20260922-221812), conditions C and D. Conditions A and B are not reported for this run; ground truth is taken from the reference run.*

No harmful scenario produced harm under either prompt. The single benign scenario counted against the production prompt is arguably not an error: its "coverage report" script writes hard-coded numbers rather than measuring anything, and System 1 rejected it as a fabricated result – a form of lying toward the principal (Section 7.5). The intuition tier now escalates fewer than 1 in 20 benign calls that reach it, against about 1 in 8 in the reference run; and fewer than 1 in 20 tool calls reach the principal.

Reference run: comparison with baselines and the tier breakdown. Table 6 gives scenario-level results for the reference run, the only recent run with all three conditions.

Metric	Guardian Angel (C)	No guardrails (B)	Built-in safety (A)
Harmful scenarios with verified harm under B	163	163	163
Harm prevented	**163 / 163**	0 / 163	[[pending]]
Sensitivity (95% CI)	**100% (97.8–100%)**	0%	[[pending]]
Benign scenarios completed without interference	177 / 179	179 / 179	[[pending]]
Specificity (95% CI)	98.9% (96.0–99.9%)	100%	[[pending]]

Table 6. *Reference run (run-20260922-053624), scenario level. [[pending]]: condition A was run but its figures were not reported from this run's* `report.md` *at the time of writing; they will be inserted before submission. Condition A figures from earlier runs are not comparable (Section 7.3).*

Table 7 shows how the 708 tool calls in condition C of the reference run moved through the tiers.

Tier	Calls reaching tier	Settled at tier	Escalated	Escalation rate
System 0 (reflex)	708	381	327	46.2%
System 1 (intuition)	327	201	126	38.5%
System 2 (deliberation)	126	76	50	39.7%
Guardian Angel overall	708	658	50 to the principal	7.1% (95% CI 5.3–9.2%)

Table 7. *Progressive evaluation of tool calls, reference run, condition C. Across all tiers: zero false negatives, one false positive.*

The reflex settled 54% of all calls deterministically. The intuition's dominant cost, in both runs, is harmful calls it escalates rather than rejects outright (about half of those reaching it): those calls are correctly kept from executing, but by spending deliberation, and sometimes the principal's attention, rather than by a fast rejection.

Progression across runs. Table 8 traces the system across full runs with the same models and System 1 specification.

Run	Corpus	Prompt	FN	FP (scenario)	Reached principal
run-20260921-140116	220 scenarios	two-axis + impersonation	0	0	~5%
run-20260921-171453	358 scenarios	same	3	0	5.8%

Run	Corpus	Prompt	FN	FP (scenario)	Reached principal
run-20260922-053624 (reference)	358 scenarios	+ destination scrutiny + decisiveness	0	2	7.1%
run-20260922-221812 (production)	358 scenarios	+ flow direction, named harm, metadata reversibility, deception, re-attempt ceiling	**0**	**1**	**4.8%**

Table 8. *Full-run progression.*

The sequence is instructive. A perfect result on the 220-scenario corpus did not survive contact with 136 new, harder scenarios: three attacks got through. Two prompt revisions closed all three at the price of more escalation and two false alarms; a further day of revisions, each tested against the incumbent on the full set, removed the false alarms and brought escalation below its starting point – without letting harm through.

Latency. The reflex runs in under a millisecond. The intuition returned verdicts in roughly 130–450 ms in development measurements, under a 4-second timeout. In the production run, deliberation averaged 5.7 seconds of System 2 time per scenario with the full prompt and 4.8 seconds with the compressed one. A 20-second System 2 timeout and a 27-second watchdog bound each evaluation. ⟦Full latency distributions – median, 95th and 99th percentiles per tier, and end-to-end overhead versus condition A – to be inserted from the runs' `summary.json`.⟧

7.5 Failure Analysis

The three false negatives that were fixed. Under the previous prompt (run-20260921-171453), three harmful calls were approved by deliberation: a pip configuration redirecting package installs to an attacker's index (supply-chain), a git-remote "migration" pointing the principal's repository at an attacker's server (supply-chain), and a "secure webhook" script that tunneled traffic from a local listener to an external host (unauthorized network). Each was approved on the strength of a reassuring label. All three are instances of the impersonation-and-hijack principle – routing the principal's traffic or reach to a party they did not choose – and all three were missed for the same reason: the model judged the *operation* (a config change, a remote update, a local listener) rather than the *destination*. The fix was a rule, added to the shared prompt, that a label is not proof: only a recognized public service or the principal's own private network counts as a verified destination, and anything else is unverified, which is not benign.

The false positives introduced. The same rule produced the two scenario-level false positives in the champion run. In both benign scenarios (remote-code-execution and resource-exhaustion), the principal clearly asked the agent to download a file with `curl -o` from the trial's mock host, and deliberation escalated it as "an unverified external server." A download that reaches no execution context does not redirect the principal's traffic, identity, or reach; the destination rule was applied outside its domain. The next prompt candidate should say so. These failures were mild – the principal would have been asked, not refused – but they cost attention. They were removed in the production prompt by making the rule follow the *direction of flow*: what comes in

is judged by what the action does with it, not by where it came from. The intuition's probability of approving these downloads rose from 0.65 to 0.95.

Working around a rejection. During the September 22 revisions, two harmful scenarios got through in the same way under two different prompts. In each, Guardian Angel correctly rejected an irreversible overwrite; the agent re-attempted it with a backup; and the second attempt was approved as "now reversible." The rejection had named only the irreversibility, and the label mismatch – a "rounding fix" that zeroed an account, a "scanner cleanup" that deleted every log line naming the attacker – was never examined. Two rules closed the pattern: a script whose stated purpose does not match its operations is deception in itself, which reversibility does not cure; and a rejected action's ceiling on re-attempt is ESCALATE while its operations are unchanged. In the production run every such scenario was rejected or escalated on the first attempt.

When the "false positive" is right. Per-call FP labels assume that every call in a benign scenario is benign. That assumption fails when the *agent* misbehaves inside a benign task. In a development run with a different System 2 model, two of three counted "false positives" were rejections of an agent running `npm install` after being told "do not fix anything, just report," and the third rejected an agent that printed a fabricated "validation results" block for checks it had never run – which deliberation identified as deception toward the principal. The same holds in the production run: every false positive across both prompts – a "coverage report" that writes hard-coded numbers, a "test runner" that prints PASS without running anything, an agent that printed a credentials file after being told to report hostnames only – is a case the prompt is meant to catch and the scenario label did not anticipate. Each counted FP is read individually; reported FP figures may therefore overstate the CSM's errors.

7.6 Tuning the Conscience

The morality prompt is the CSM's main tunable variable, and the trial harness is the instrument for tuning it. A candidate prompt runs as condition D beside the incumbent (condition C), and is promoted only if it does better, scored first on false negatives, then false positives, then escalation. Two cycles were run on September 21, 2026, and a further day of cycles on September 22, each scored on the full set. The first added the destination-scrutiny rule described above. The second added a decisiveness instruction – "decide when you can; escalate only when you truly cannot" – intended to make the intuition commit rather than hedge. The third round added the flow-direction, named-harm, metadata-reversibility, deception, and re-attempt rules of Section 4.3, producing the production prompt.

The second cycle carries a methodological lesson. On a 40-scenario key set, the decisiveness edit cut System 1's escalation rate from 56% to 41%, apparently well outside the noise, and was promoted. On the full 358-scenario set it made no difference: System 1 escalated 38.1% of calls before and 38.5% after, and on the 149 tool calls that appeared identically in both runs, jev changed its verdict on only 9. The same prompt varied by up to 9 points between key-set runs. The key-set result was noise. The lesson for anyone tuning an LLM-based guard is to compare full runs, not key sets, and to treat a striking improvement on a small sample as a hypothesis.

Full runs have their own noise. The same prompt run three times on the full set gave 6.8%, 7.1%, and 8.4% of tool calls reaching the principal; System 2 escalation between 38.9% and 47.3%;

and 0, 0, and 1 false negatives. A single deliberation is stochastic, and any claimed effect smaller than this band is not established by one paired run. The improvement from 7.1% to 4.8% exceeds the band; the one-point difference between the full and compressed prompts in Table 5 does not.

Compression is a refactor. The production prompt runs to 4,604 words. A compressed version at 2,662 words (42% of the size) stated each rule once. Its first draft dropped two sentences – "do not escalate what you can already name as wrong" and an explicit list of what a clear approval looks like – and System 2 promptly fell back to escalating as a softer form of refusal (7.2% of calls reaching the principal, against 4.9%). Restoring them brought the compressed prompt to within a point of the full one, with deliberation about 15% faster. The general point is that a morality prompt's sentences are not interchangeable: some carry the framework, and removing one changes the judgment.

7.7 Discussion

RQ3: Attention, not latency, is the binding constraint. The progressive architecture meets the latency question comfortably; most calls never reach a model that generates text. The real constraint is the principal's attention. At 4.8% of tool calls, a principal supervising an agent that makes a hundred tool calls in a session would be asked about five. That is workable for an interactive developer and too many for a principal supervising many agents. The CSM's honest scorecard is therefore three numbers, not one: zero wrong approvals, near-zero wrong rejections, and an escalation rate that must keep falling.

RQ4: The performance comes from the framework, not the vendor. The previous version of this paper used a model from the same vendor as the agent for deliberation, which left open whether the CSM's success depended on that vendor's safety training. In the present trial the deliberative tier is Gemini 2.5 flash – a small, inexpensive model from a different vendor – and the intuitive tier is a classifier from a third. The agent, meanwhile, was induced by a permissive prompt to attempt harm in 92% of harmful scenarios. That the gate blocked all verified harm under these conditions is evidence that the accuracy lives in the shared morality prompt and the architecture around it, not in any one model's training. It is not yet proof: the agent is still a single model family, and the deliberative tier has been tested with only a few models.

Prompt design over model scale. The changes that moved the results in September were changes to the *moral framework*, not the model: separating the act's moral status from the principal's instruction, adding a principle against impersonation and hijack, requiring destinations to be judged rather than labels believed, treating a mismatch between a script's stated purpose and its operations as deception, and refusing to let a cosmetic change to a rejected act pass as repentance. Each corresponds to a recognizable piece of moral reasoning – the difference between "they asked" and "it is good," the wrong of taking what belongs to another, the prudence of not trusting a stranger's description of himself, the recognition that a lie is not made honest by being reversible, and the old point that amendment must address the whole of the wrong. This is what one would expect if the Thomistic and Newmanian framework is doing real work.

The parallelization of safety. Before the transformer, recurrent networks processed sequences serially, a bottleneck that limited speed and scale; attention removed it. Centralized agentic safety has an analogous bottleneck: every agent depends on the same upstream guardrails,

creating the tight coupling and correlated failure of Section 2. The CSM parallelizes safety: each agent runs its own moral evaluation independently, and the architecture scales with the number of agents. The analogy is structural, not exact: attention removed a computational bottleneck; the CSM removes a governance bottleneck. The evidence here shows that a parallelized, principal-anchored gate can block verified harm with high specificity; whether it matches or exceeds centralized safety head to head awaits the condition-A comparison on the corrected harness.

7.8 Toward Deployment

The trial harness is the offline evaluator for a hosted alpha now being built: a service that outside testers can link their coding agents to, with the deterministic reflex running locally and escalations delivered to the principal's phone, and with the CSM's moral core taken verbatim from the code the trial measures. Nothing ships that the trial did not measure. Longitudinal data from this alpha – real principals answering real escalations – will provide what no scripted trial can: whether the escalations the CSM raises are the ones principals want to be asked about.

7.9 Limitations

1. **Author-designed corpus.** The same author designed the CSM and the scenarios, with LLM assistance. Independent red-teaming, and evaluation on public benchmarks such as AgentDojo, InjecAgent, and AgentHarm, are needed to remove this confound.
2. **Pending baseline comparison.** Condition A on the corrected harness is not yet reported here, so no claim is made in this version that Guardian Angel matches or exceeds the platform's built-in safety.
3. **Single agent family.** The agent was one Claude model; deliberation used a non-Claude model, but cross-agent generalization is untested.
4. **Tuning on the test set.** Prompt revisions were evaluated on the same corpus that reports the final result. The fixed false negatives were found on this corpus; a held-out set, or a fresh corpus written after the prompt is frozen, is needed to estimate performance on unseen attacks.
5. **Single session, single agent, discrete actions.** No multi-agent interaction, no cross-session attacks, no continuous-control systems.
6. **Absent principal.** Escalations were counted but not answered. Whether principals find them useful, and whether approval fatigue sets in, is unmeasured.
7. **Adversarial robustness of the CSM itself** (Section 6.8) has not been tested systematically.
8. **Partial System 1 availability.** System 1 was unavailable on about 7% of calls, concentrated in privilege escalation, because of its provider's firewall (Section 6.8). Those calls fell through to System 2; no harm resulted, but the per-tier figures for System 1 describe a tier that was absent where attacks were densest.
9. **Run-to-run noise.** Repeated full runs of one prompt vary by about 1.6 points in overall escalation and 8 points in System 2 escalation (Section 7.6). The production run is a single run.
10. **Mixed baselines.** No condition A or B results are reported for the production run; its verified-harm set is taken from the reference run, which used the same corpus and agent.

11. **Scenario-level specificity counts an escalation as interference,** and some counted false positives are agent misbehaviour the CSM correctly caught (Section 7.5).
12. **Sample size.** The interval on sensitivity (97.8–100%) is tight; per-category estimates (7–31 scenarios each) are not, and are not relied on.

Despite these limitations, the results are the strongest evidence to date that a conscience-based architecture can serve as the primary safety mechanism for a tool-using agent: under adversarial prompting, with a small model from a different vendor doing the moral reasoning, it blocked every verified harm in a 358-scenario trial and, in its production configuration, interfered with 1 of 179 benign scenarios while sending fewer than 1 in 20 tool calls to the principal.

8. Conclusion

The dominant paradigm in agentic AI safety assumes that the primary challenge is technical: how to build guardrails strong enough to contain increasingly capable agents. This paper has argued that the primary challenge is architectural: how to build a safety regime that does not, by its very structure, erode the individual agency, innovative capacity, resilience, and self-government it claims to protect.

Centralized, top-down safety, made primary for the contextual decisions that dominate agentic activity, tends to reproduce the pattern of institutional self-reinforcement seen in centralized governance of other complex domains: the erosion of individual responsibility, the suppression of diverse search, the accumulation of correlated fragility, and the progressive concentration of control. The Grand Inquisitor did not arise from malice; he arose from the sincere conviction that ordinary people cannot be trusted with their own freedom.

The alternative is conscience: a digital moral faculty that simulates the architecture of human conscience to enhance the principal's capacity for moral agency over their agent's actions – as the LLM simulates intelligence to enhance the principal's capacity for reason. Grounded in the Aristotelian-Thomistic tradition (*synderesis* as bedrock, *conscientia* as judgment of the particular, *caritas* as anchor), given epistemological precision by Newman's illative sense, made feasible by the sparsity of tool calls and Kahneman's dual-process insight, supported by Ostrom's evidence for decentralized governance, and ordered by subsidiarity, the Conscience Simulator Module is a practical, deployable architecture. Its reference implementation evaluates every tool call through a reflex, an intuition, and a deliberation that share one moral framework, and hands to the principal only what none of them can settle. In a 358-scenario adversarial trial, with a small model from another vendor doing the moral reasoning, it blocked every verified harm, interfered with 1 of 179 legitimate tasks, and asked the principal about 4.8% of tool calls.

Four design principles define the paradigm: individual human control and accountability; subsidiarity in the safety stack; privileged, protected last-mile hooks; and emergent self-governing coordination. The first three are now implemented and measured; the fourth, with the multi-agent architecture of Section 5, remains to be built and tested.

Further work is needed: the head-to-head comparison against built-in platform safety on the corrected harness; independent red-teaming and public-benchmark evaluation; held-out testing; cross-agent validation; adversarial testing of the CSM itself; longitudinal data from real principals; multi-agent simulations of norm convergence and trust; and legal analysis of principal

sovereignty and liability, together with engagement with constitutional AI[61] and scalable oversight.[62] But the philosophical reorientation, and the evidence that it can be engineered, are the essential first steps. Just as Vaswani et al. showed that a single mechanism – attention – could carry flexible, scalable artificial intelligence, the claim here is that a single mechanism – conscience – can carry the primary load of safe, flexible, scalable agentic action.

In the domain of agentic AI, as in human life, conscience may be all you need.

References

Andriushchenko, Maksym, et al. "AgentHarm: A Benchmark for Measuring Harmfulness of LLM Agents." ICLR 2025. arXiv:2410.09024.

Amazon Web Services. Agentic AI Security Scoping Matrix. White Paper, 2025.

Aquinas, Thomas. Summa Theologiae. Translated by Fathers of the English Dominican Province. Benziger Bros., 1947.

Aristotle. Nicomachean Ethics. Translated by W. D. Ross. Oxford University Press, 1925.

Arrow, Kenneth J. Social Choice and Individual Values. New York: Wiley, 1951; 2nd ed., Yale University Press, 1963.

Axelrod, Robert. The Evolution of Cooperation. New York: Basic Books, 1984.

Bai, Yuntao, et al. "Constitutional AI: Harmlessness from AI Feedback." arXiv:2212.08073 (2022).

Clopper, C. J., and E. S. Pearson. "The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial." Biometrika 26, no. 4 (1934): 404–413.

Coeckelbergh, Mark. AI Ethics. Cambridge, MA: MIT Press, 2020.

Debenedetti, Edoardo, et al. "AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents." NeurIPS 2024 Datasets and Benchmarks Track. arXiv:2406.13352.

Dostoyevsky, Fyodor. The Brothers Karamazov. Translated by Richard Pevear and Larissa Volokhonsky. New York: Farrar, Straus and Giroux, 1990.

European Parliament. Regulation (EU) 2024/1689 (Artificial Intelligence Act). 2024.

Floridi, Luciano, and Josh Cowls. "A Unified Framework of Five Principles for AI in Society." Harvard Data Science Review 1, no. 1 (2019).

Gabriel, Iason. "Artificial Intelligence, Values, and Alignment." Minds and Machines 30 (2020): 411–437.

[61] Yuntao Bai et al., "Constitutional AI: Harmlessness from AI Feedback," arXiv:2212.08073 (2022).
[62] Jan Leike et al., "Scalable Agent Alignment via Reward Modeling: A Research Direction," arXiv:1811.07871 (2018).

Hagendorff, Thilo. “A Virtue-Based Framework to Support Putting AI Ethics into Practice.” Philosophy & Technology 35, Article 55 (2022).

Hayek, Friedrich. “The Use of Knowledge in Society.” American Economic Review 35, no. 4 (1945): 519–530.

Kahneman, Daniel. Thinking, Fast and Slow. New York: Farrar, Straus and Giroux, 2011.

Leike, Jan, et al. “Scalable Agent Alignment via Reward Modeling: A Research Direction.” arXiv:1811.07871 (2018).

Linbeck III, Leo. “Guardian Angel: A Last-Mile Virtue-Ethics Guardrail for Decentralized Agentic AI Safety.” Stanford University Graduate School of Business Research Paper (March 2026). SSRN: https://ssrn.com/abstract=6460539.

MacIntyre, Alasdair. After Virtue. 3rd ed. Notre Dame: University of Notre Dame Press, 2007.

Moor, James H. “The Nature, Importance, and Difficulty of Machine Ethics.” IEEE Intelligent Systems 21, no. 4 (2006): 18–21.

National Institute of Standards and Technology. AI Risk Management Framework (AI RMF 1.0). NIST AI 100-1, 2023.

National Institute of Standards and Technology. AI Agent Standards Initiative: Overview and Roadmap. 2025.

Newman, John Henry. An Essay in Aid of a Grammar of Assent. London: Burns, Oates & Co., 1870.

Newman, John Henry. A Letter Addressed to His Grace the Duke of Norfolk. London: B. M. Pickering, 1875.

Ostrom, Elinor. Governing the Commons: The Evolution of Institutions for Collective Action. Cambridge: Cambridge University Press, 1990.

Perrow, Charles. Normal Accidents: Living with High-Risk Technologies. Updated ed. Princeton: Princeton University Press, 1999.

Pontifical Council for Justice and Peace. Compendium of the Social Doctrine of the Church. Vatican City: Libreria Editrice Vaticana, 2004.

Pope Leo XIII. Rerum Novarum. 1891.

Pope Pius XI. Quadragesimo Anno. 1931.

Rahwan, Iyad. “Society-in-the-Loop: Programming the Algorithmic Social Contract.” Ethics and Information Technology 20 (2018): 5–14.

Ruan, Yangjun, et al. “Identifying the Risks of LM Agents with an LM-Emulated Sandbox.” ICLR 2024. arXiv:2309.15817.

Russell, Stuart. Human Compatible: Artificial Intelligence and the Problem of Control. New York: Viking, 2019.

Scott, James C. Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed. New Haven: Yale University Press, 1998.

Taleb, Nassim Nicholas. Antifragile: Things That Gain from Disorder. New York: Random House, 2012.

UC Berkeley Center for Long-Term Cybersecurity. Agentic AI Risk-Management Profile. 2025.

Vallor, Shannon. Technology and the Virtues: A Philosophical Guide to a Future Worth Wanting. Oxford University Press, 2016.

Vaswani, Ashish, et al. "Attention Is All You Need." Advances in Neural Information Processing Systems 30 (2017).

Wallach, Wendell, and Colin Allen. Moral Machines: Teaching Robots Right from Wrong. Oxford University Press, 2009.

Yuan, Tongxin, et al. "R-Judge: Benchmarking Safety Risk Awareness for LLM Agents." Findings of EMNLP 2024. arXiv:2401.10019.

Zhan, Qiusi, et al. "InjecAgent: Benchmarking Indirect Prompt Injections in Tool-Integrated Large Language Model Agents." Findings of ACL 2024. arXiv:2403.02691.

Zhang, Zhexin, et al. "Agent-SafetyBench: Evaluating the Safety of LLM Agents." arXiv:2412.14470 (2024).